

Signals and Systems Course Project Report

Generated project draft

2026 Spring

1 Project Objective

This project studies basic signal transformations and frequency-domain analysis using a real audio signal. The sampled audio is denoted by $x[n]$. Using interpolation, it is treated as an approximation of a continuous-time signal $x(t)$. The project then analyzes time reversal, time scaling, Fourier spectra, magnitude-only and phase-only reconstruction, and ideal low-pass filtering.

2 Input Audio and MATLAB Environment

The implementation language is MATLAB. The input audio file is `audio/input/original.wav`. It has the following properties:

- Format: WAV, PCM signed 16-bit little-endian.
- Sampling rate: 16000 Hz.
- Channels: 1 channel, mono.
- Duration: 7.000 seconds.
- Preprocessing: amplitude normalization to avoid clipping.

The audio is short enough for clear waveform and spectrum plots, but long enough to show meaningful frequency content.

3 Interpolation and Continuous-Time Counterpart

The discrete audio samples $x[n]$ are read by MATLAB using `audioread`. To construct a computable approximation of $x(t)$, the project uses linear interpolation between adjacent samples. In MATLAB this is implemented with `interp1`. Figure 1 shows the sampled values and the interpolated curve over a short time window.

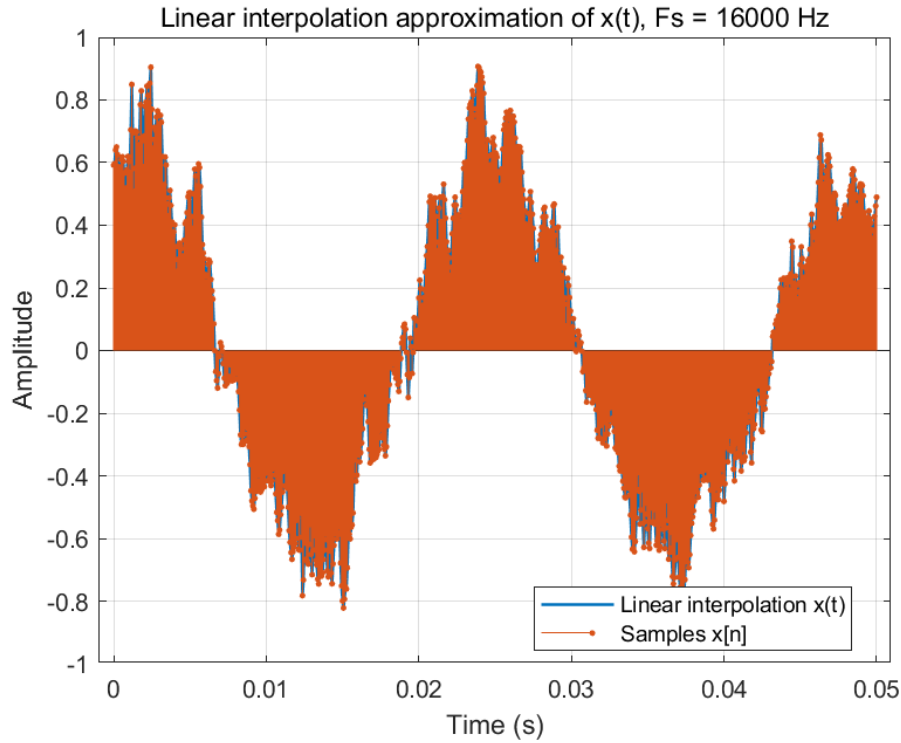


Figure 1: Linear interpolation approximation of $x(t)$.

4 Time-Domain Transformations

Three time-domain transformations are generated:

- $x(-t)$: time reversal, implemented by reversing the audio samples.
- $x(2t)$: time compression by a factor of 2. The resulting duration is about 3.5 seconds.
- $x(t/2)$: time expansion by a factor of 2. The resulting duration is about 14 seconds.

The time-reversed signal sounds like the original clip played backward. The compressed signal changes faster in time, while the expanded signal changes more slowly.

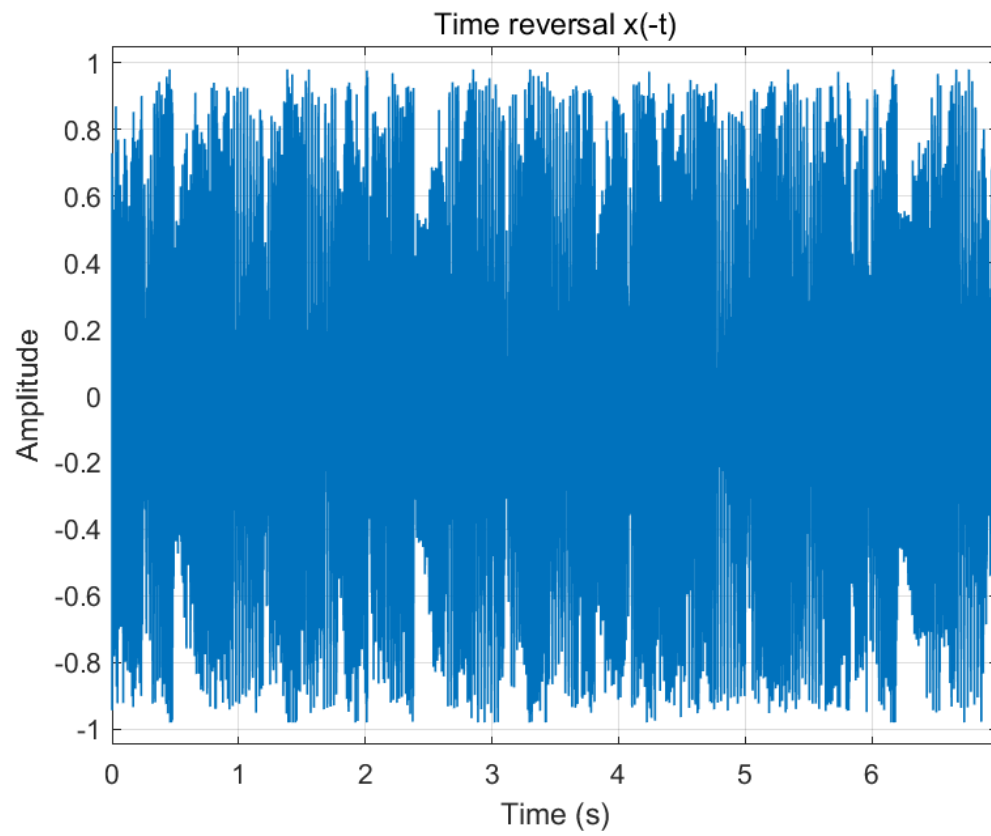


Figure 2: Waveform of $x(-t)$.

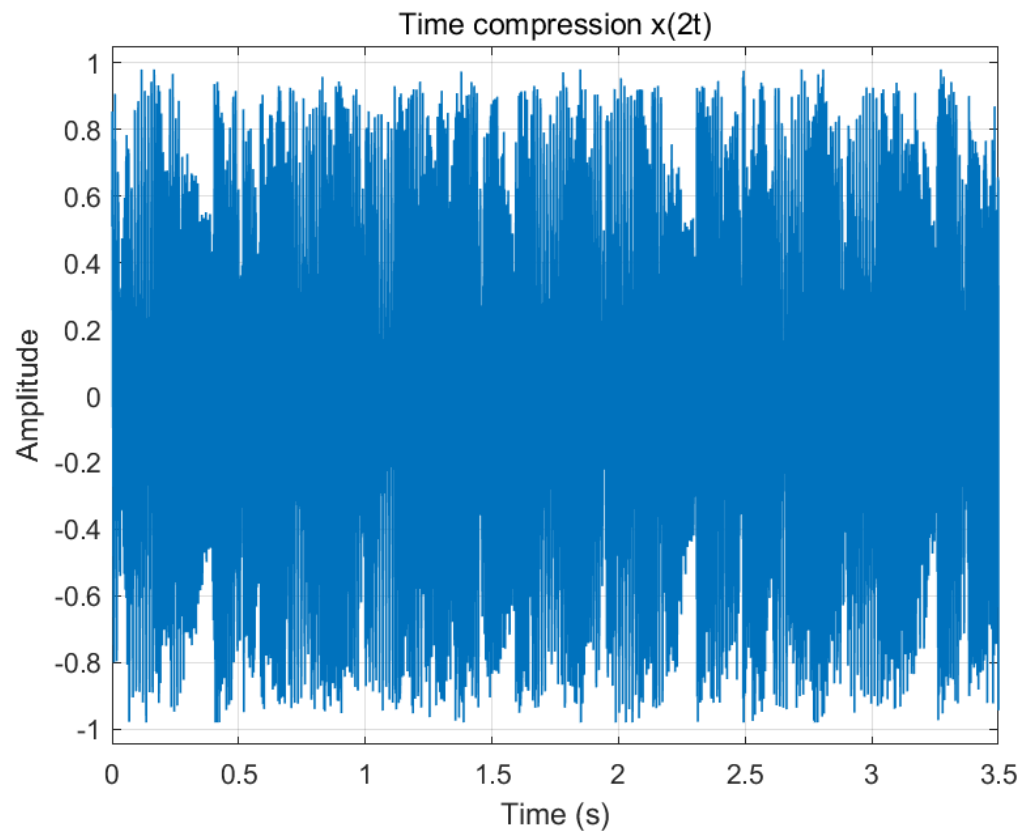


Figure 3: Waveform of $x(2t)$.

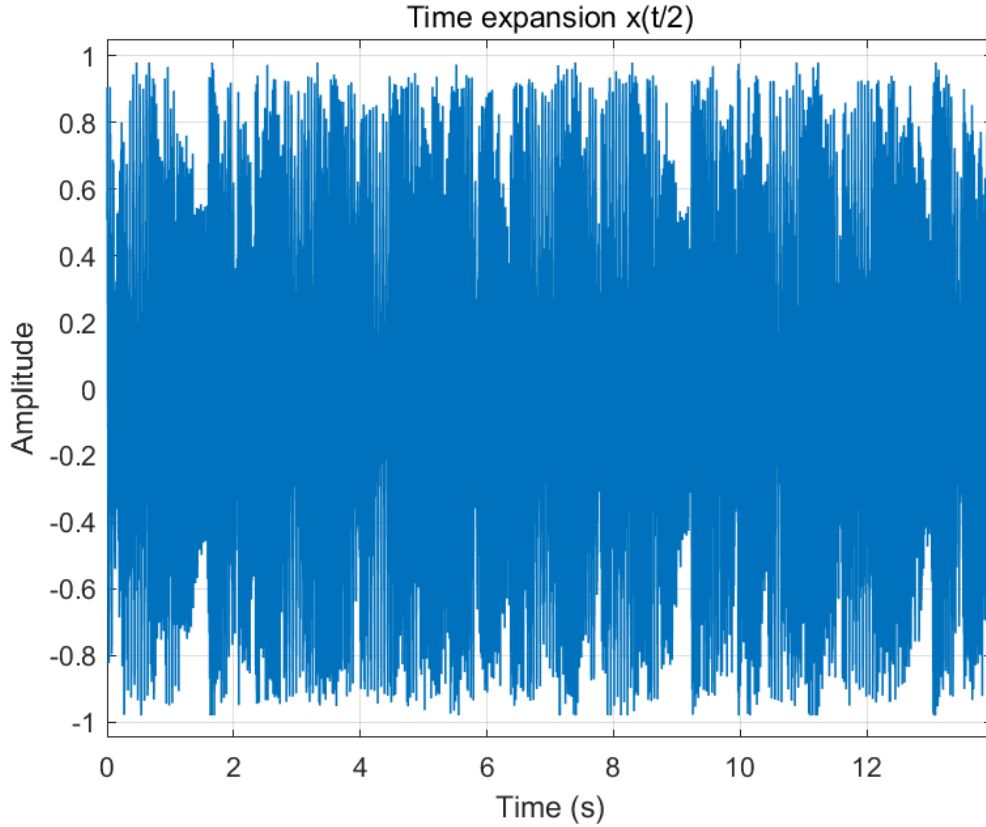


Figure 4: Waveform of $x(t/2)$.

5 Fourier Transform and Spectrum Analysis

The Fourier spectra are computed using the FFT. For visualization, `fftshift` places negative frequencies on the left and positive frequencies on the right. The magnitude spectra are normalized and plotted in dB.

For time reversal, the magnitude spectrum is largely preserved, while phase is changed. For $x(2t)$, time compression corresponds to frequency-axis expansion. For $x(t/2)$, time expansion corresponds to frequency-axis compression. These observations agree with the Fourier transform time-scaling property:

$$x(at) \iff \frac{1}{|a|} X\left(j\frac{\omega}{a}\right).$$

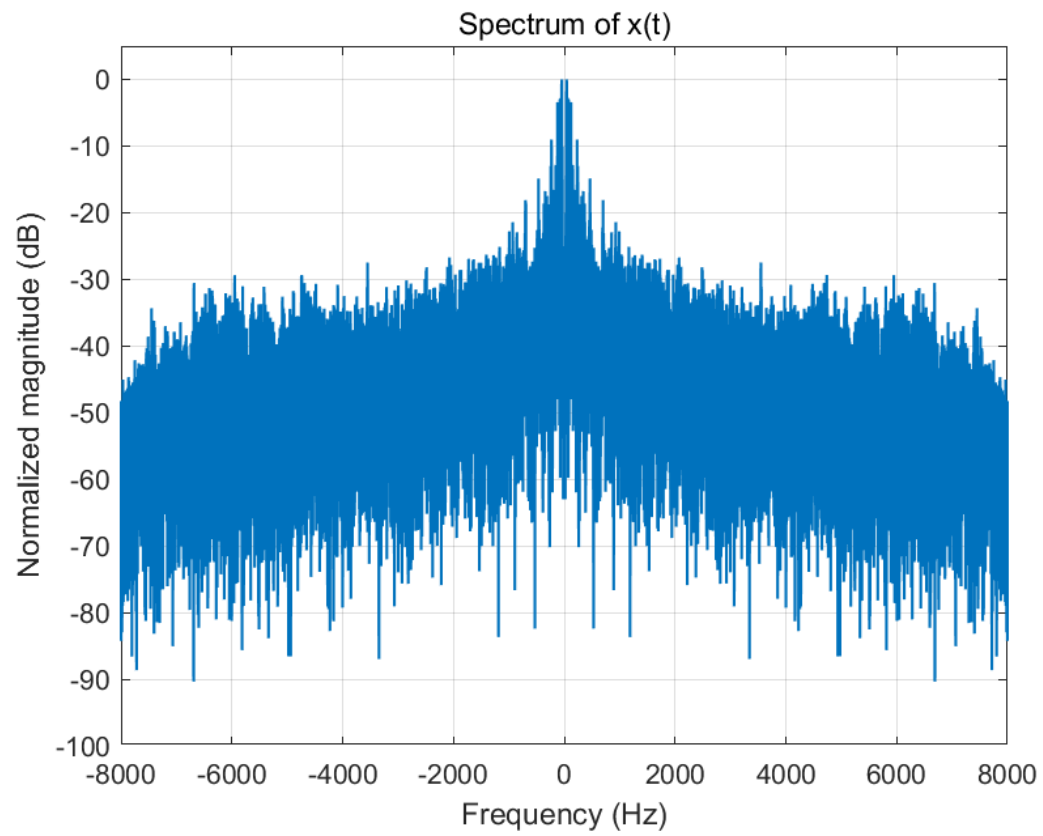


Figure 5: Spectrum of the original signal.

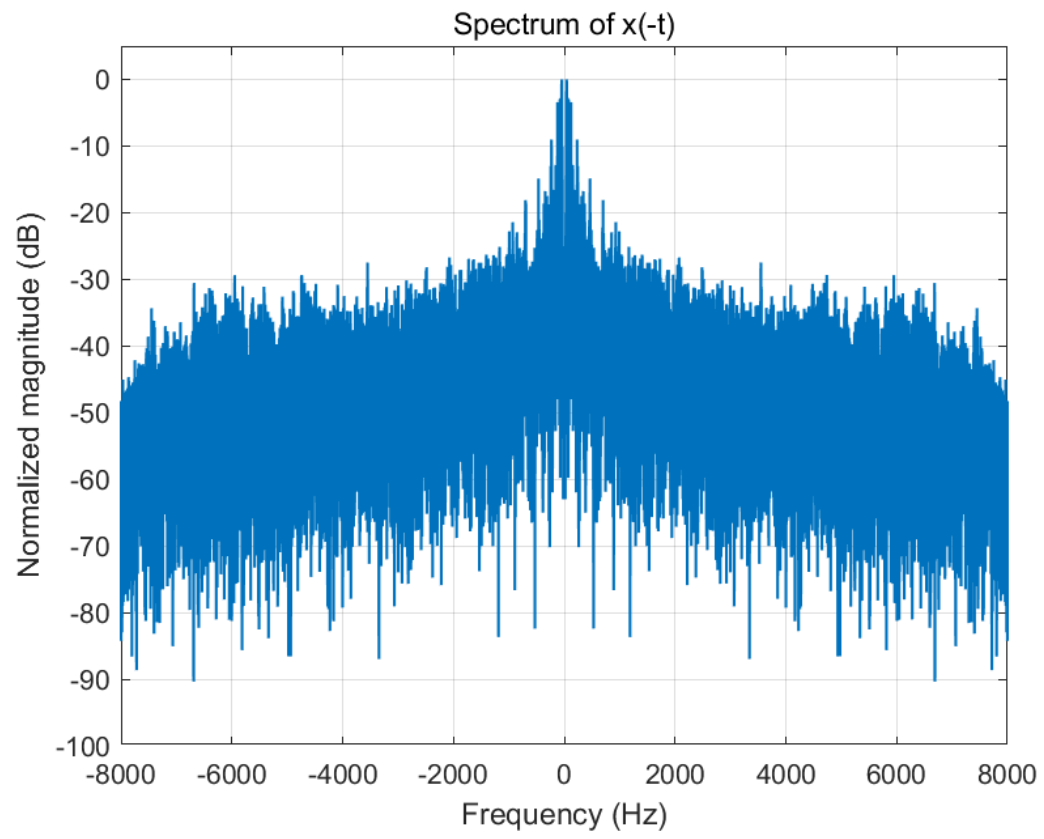


Figure 6: Spectrum of $x(-t)$.

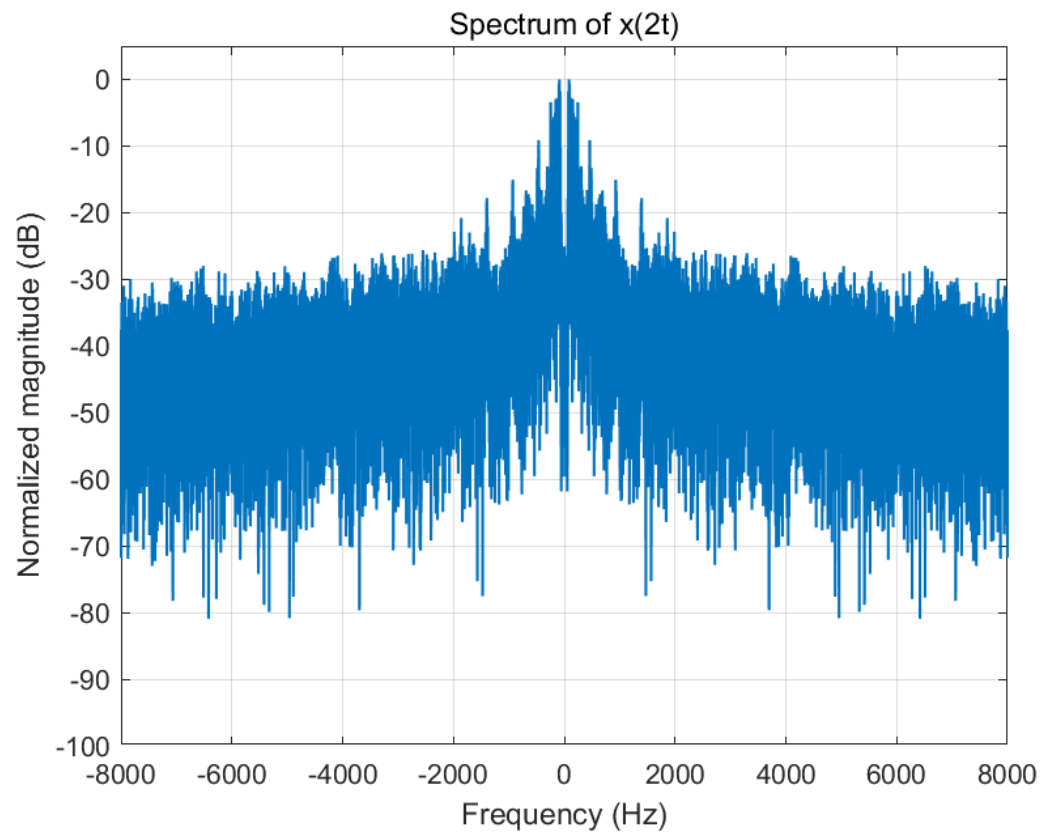


Figure 7: Spectrum of $x(2t)$.

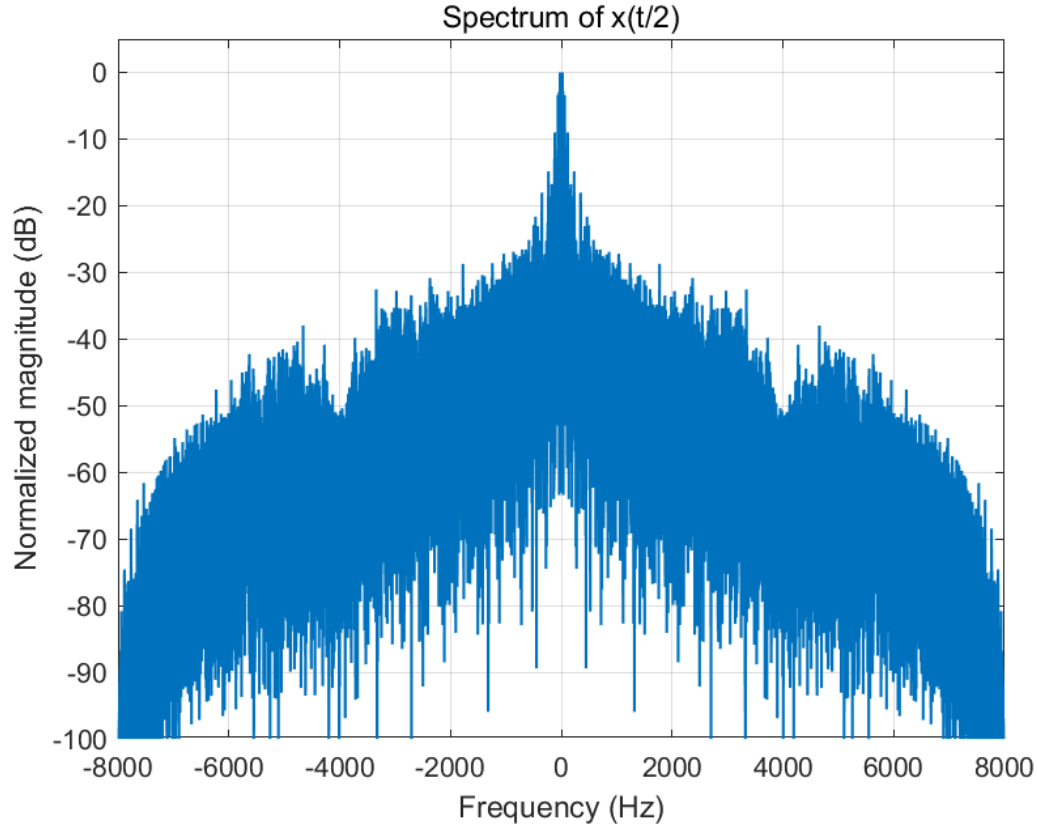


Figure 8: Spectrum of $x(t/2)$.

6 Magnitude-Only and Phase-Only Reconstruction

Let $X(j\omega)$ be the Fourier transform of $x(t)$. The project compares:

$$\mathcal{F}^{-1}\{|X(j\omega)|\}$$

and

$$\mathcal{F}^{-1}\{e^{j\angle X(j\omega)}\}.$$

The magnitude-only reconstruction keeps the spectral energy distribution but throws away phase. The phase-only reconstruction keeps timing and phase relationships but discards the original magnitude distribution. The resulting waveforms are very different from the original, showing that both magnitude and phase are important for reconstructing an audio signal.

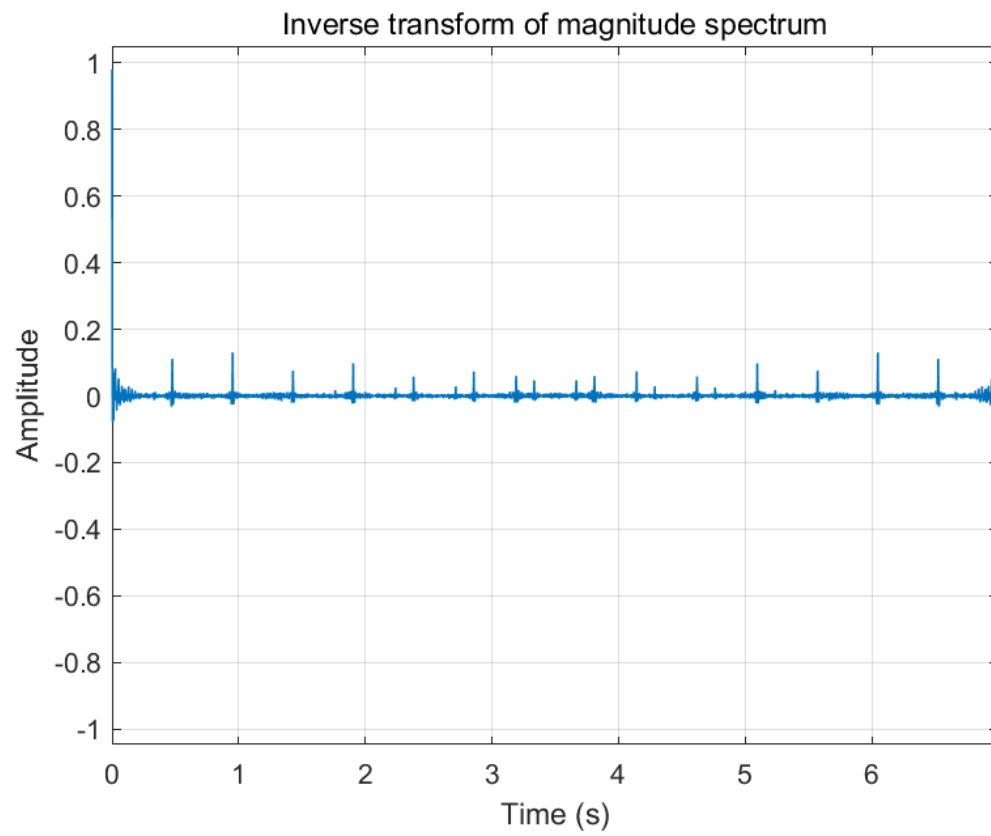


Figure 9: Inverse transform of the magnitude spectrum.

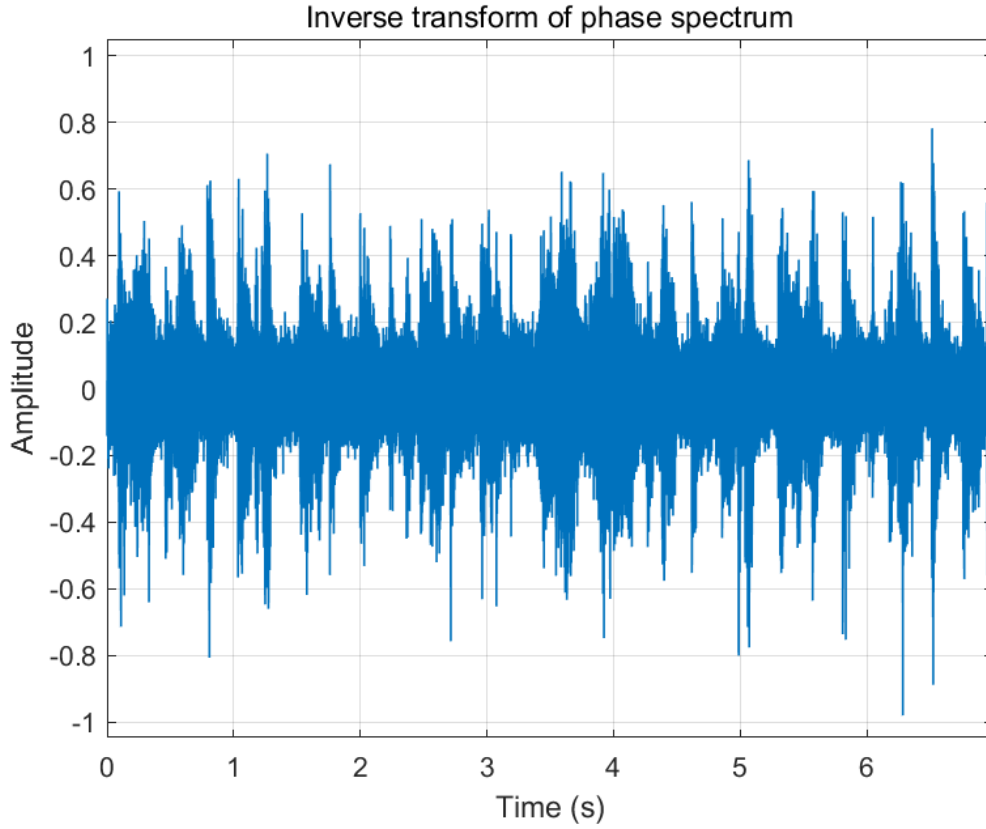


Figure 10: Inverse transform of the phase spectrum.

7 Frequency-Domain Low-Pass Filtering

An ideal low-pass filter is applied in the frequency domain. Since the sampling rate is 16000 Hz, the Nyquist frequency is 8000 Hz. The chosen cutoff frequency is 3000 Hz. The filter response is

$$H(f) = \begin{cases} 1, & |f| \leq 3000 \text{ Hz}, \\ 0, & |f| > 3000 \text{ Hz}. \end{cases}$$

This filter removes high-frequency components while preserving the low and middle audio band. The filtered waveform and spectrum are shown below.

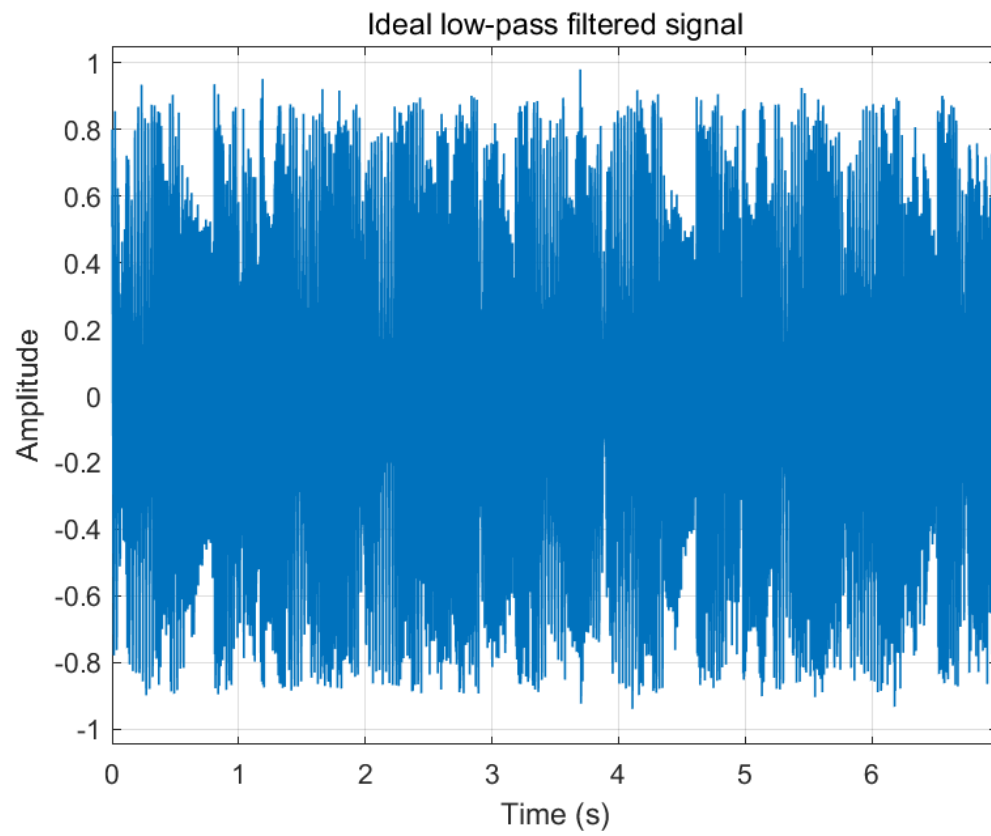


Figure 11: Waveform after ideal low-pass filtering.

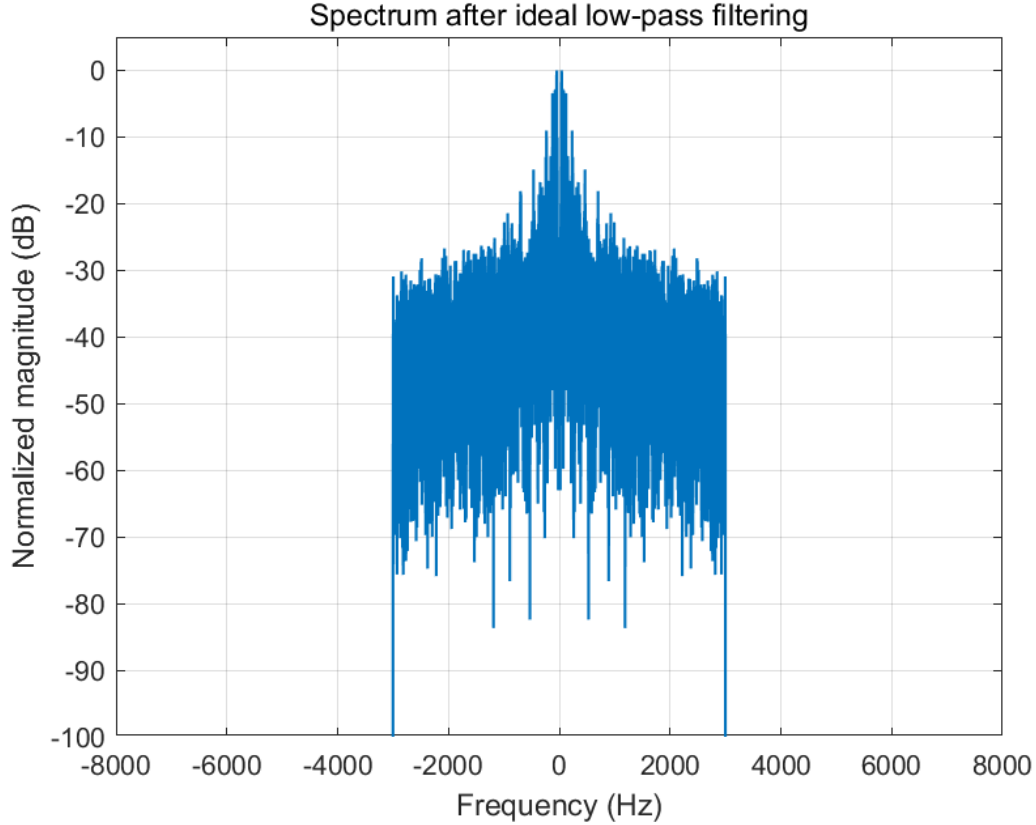


Figure 12: Spectrum after ideal low-pass filtering.

8 Part II: OTFS Paper Study

The selected paper is *Interference Cancellation and Iterative Detection for Orthogonal Time Frequency Space Modulation* by P. Raviteja, Khoa T. Phan, Yi Hong, and Emanuele Viterbo. It was published in IEEE Transactions on Wireless Communications in 2018. A public preprint is available at <https://arxiv.org/abs/1802.05242>.

The paper studies OTFS modulation over delay-Doppler channels. In high-mobility scenarios, OFDM can suffer from Doppler spread, which breaks subcarrier orthogonality and creates inter-carrier interference. OTFS addresses this problem by placing information symbols in the delay-Doppler domain and then mapping them to the time-frequency domain for transmission.

The main idea can be translated and summarized as follows. A wireless channel with multipath delay and Doppler shift has a natural sparse structure in the delay-Doppler domain. Instead of observing the channel as rapidly changing subcarriers, OTFS observes it through a two-dimensional representation connected to physical path delays and velocities. The authors derive an input-output relationship for OTFS modulation and demodulation, analyze the resulting interference, and propose a message-passing detector that iteratively cancels interference and estimates the transmitted symbols.

OTFS can overcome frequency-selective fading because multipath delays are represented explicitly along the delay dimension. It can also handle fast mobility because Doppler shifts are represented explicitly along the Doppler dimension. Since practical wireless channels often contain only a limited number of dominant propagation paths, the delay-Doppler representation is sparse. This sparsity makes joint detection and interference cancellation practical. Therefore, OTFS is more robust than conventional OFDM in channels with severe time variation and Doppler spread.

9 Generated Files

File	Meaning
audio/output/original_normalized.wav	Normalized source audio
audio/output/x_reverse.wav	Time reversal $x(-t)$
audio/output/x_2t.wav	Time compression $x(2t)$
audio/output/x_t_over_2.wav	Time expansion $x(t/2)$
audio/output/magnitude_only_reconstruction.wav	Magnitude-only reconstruction
audio/output/phase_only_reconstruction.wav	Phase-only reconstruction
audio/output/lowpass_ideal_3000Hz.wav	Ideal low-pass result

Table 1: Generated audio files.

10 Conclusion

The MATLAB implementation completes all signal-processing requirements of the project. The experiments demonstrate how time-domain transformations affect both waveform duration and spectral distribution, why magnitude and phase are both necessary for faithful reconstruction, and how frequency-domain filtering can be implemented through the Fourier transform. The OTFS paper study further connects transform-domain signal characterization with modern wireless communication systems.